An Analysis of the Kleinberg-Tardos MaxSAT Approximation Algorithm

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The simple randomized algorithm presented in Chapter 13.4 of Kleinberg-Tardos[1] is one of the simplest for approximating SAT, yet is also surprising effective, satisfying on average over 7/8 of clauses for 3-CNF instances. In this paper we extend the analysis of this algorithm (henceforth the "KT algorithm") and explore two research questions:

- What is the distribution of satisfied clauses if the number of literals in a clause is random?
- What is the distribution of satisfied clauses relative to (1) the proportion of variables assigned as true/false and (2) proportion of positive literals in each clause?

0 - Algorithm Review

The algorithm works by randomly assigning either `True` or `False` to variables. If we let $x_1, \ldots, x_n$ be the set of variables in a formula, the KT algorithm assigns Bernoulli variables $b_1, \ldots, b_n$ to the variables in order to build a model M where

$$M(x_i) = \begin{cases} \text{True} & b_i = 1 \\ \text{False} & b_i = 0 \end{cases}$$

The discussion in Kleinberg-Tardos assumes that $P(b_i = 1) = 1/2$.

1 - Random number of literals

Given a CNF formula f with m clauses and X is a random variable that stands for the proportions of the clauses in f satisfied by the KT algorithm, we know from a previous analysis that

[1] Jon Kleinberg and Éva Tardos, *Algorithm Design* (Pearson, 2005).

$$\mathrm{E}\left[X_f\right] = 1 - \frac{1}{2^l}$$

where l is the constant number of literals per clause. What happens when the number of literals per clause is random?

Let $L_1, \ldots, L_m$ be random variables that stand for the number of literals in a clause. A natural distribution to assume for the number of literals is

$$\forall_{1 \leq i \leq m}.\, L_i \sim \mathrm{Uniform}(1, n)$$

for some $n > 1$. Then what is $\mathrm{E}\left[X_f\right]$?

Let $C = \{C_1, \ldots, C_m\}$ be indicator variables such that $C_i = 1$ if clause i is satisfied, and $C_i = 0$ otherwise. If we let Y_{ij} be an indicator variable where $Y_{ij} = 1$ if the jth literal in clause i is true, and $Y_{ij} = 0$ otherwise, then we know that for any $C_i \in C$,

$$\mathrm{E}\left[C_i \mid L_i = l_i\right] = P(C_i = 1 \mid L_i = l_i) = 1 - P\left(\bigcap_{j=1}^{l_i} Y_{ij} = 0\right).$$

Since the set of Y_{ij} for C_i are mutually independent, and we know that $P(Y_{ij} = 1) = \frac{1}{2}$, then

$$P\left(\bigcap_{j=1}^{l_i} Y_{ij} = 0\right) = \prod_{j=1}^{l_i} P(Y_{ij} = 0) = \prod_{j=1}^{l_i} \frac{1}{2} = \frac{1}{2^{l_i}}$$

and

$$\mathrm{E}\left[C_i \mid L_i = l_i\right] = 1 - \frac{1}{2^{l_i}}.$$

By definition of X_f, we know that

$$X_f = \frac{C_1 + \ldots + C_m}{m}$$

and given that $C_1 + \ldots + C_m$ is L-measurable where $L = \{L_1, \ldots, L_m\}$, X_f is also L-measurable such that

$$\mathrm{E}\left[X_f \mid L\right] = \mathrm{E}\left[\frac{1}{m}\left(C_1 + \ldots + C_m\right) \mid L\right] = \frac{1}{m}\sum_{i=1}^{m} \mathrm{E}\left[C_i \mid L\right].$$

Since C_i is conditionally independent to all L_j where $j \neq i$,

$$\frac{1}{m}\sum_{i=1}^{m} \mathrm{E}\left[C_i \mid L\right] = \frac{1}{m}\sum_{i=1}^{m} \mathrm{E}\left[C_i \mid L_i\right]$$

and thus

$$\mathrm{E}\left[X_f\right] = \mathrm{E}\left[\mathrm{E}\left[X_f \mid L\right]\right] = \mathrm{E}\left[\frac{1}{m}\sum_{i=1}^{m} \mathrm{E}\left[C_i \mid L_i\right]\right] = \mathrm{E}\left[\frac{1}{m}\sum_{i=1}^{m}\left(1 - \frac{1}{2^{L_i}}\right)\right].$$

Since all $L_i \in L$ are uniformly distributed, by definition of expectation we know that

$$= \frac{1}{m}\sum_{i=1}^{m}\left[1 - \mathrm{E}\left[\frac{1}{2^{L_i}}\right]\right] = \frac{1}{m}\sum_{i=1}^{m}\left[1 - \sum_{j=1}^{n}\frac{1}{n}\left(\frac{1}{2^j}\right)\right].$$

Since we know the closed form of a partial sum for a geometric series,

$$= \frac{1}{m}\sum_{i=1}^{m}\left[1 - \frac{1}{n}\left(\frac{1}{2}\right)\left(\frac{1 - 1/2^n}{1 - 1/2}\right)\right] = \frac{1}{m}\sum_{i=1}^{m}\left[1 - \frac{1}{n}\left(1 - \frac{1}{2^n}\right)\right]$$

$$= \frac{1}{m}\left[m\left(1 - \frac{1}{n}\left(1 - \frac{1}{2^n}\right)\right)\right] = 1 - \frac{1}{n}\left(1 - \frac{1}{2^n}\right).$$

Notice that this is a generalization of the previous analysis for a fixed number of literals in each clause. If we let $L_i = l$ for all $L_i \in L$, then

$$\mathrm{E}\left[X_f\right] = \mathrm{E}\left[\frac{1}{m}\sum_{i=1}^{m} \mathrm{E}\left[C_i \mid L_i = l\right]\right] = \mathrm{E}\left[\frac{1}{m}\sum_{i=1}^{m}\left(1 - \frac{1}{2^l}\right)\right] = 1 - \frac{1}{2^l}.$$

which is exactly the result of the previous analysis.

Example. If $n = 3$ then

$$E\left[X_f\right] = 1 - \frac{1}{3}\left(1 - \frac{1}{2^3}\right)$$
$$= 1 - \frac{1}{3}\left(\frac{7}{8}\right) = 1 - \frac{7}{24} = \frac{17}{24}.$$

Compare this to the expected value for Max-3SAT instances (i.e., when $L_i = 3$ for all $L_i \in L$):

$$\mathrm{E}\left[X_f\right] = 1 - \frac{1}{2^3} = \frac{7}{8} = \frac{21}{24}.$$

This makes sense intuitively, since clauses are "easier" (more likely) to be satisfied when they have more clauses, and Max-3SAT instances are likely to have more literals.

2 - Proportion of Assignments and Literals

Previous analyses of the KT algorithm made two assumptions:

- The algorithm uniformly assigns `true` or `false` to variables. That is, if p equals the probability that a variable is assigned to `true`, previous analyses assumed $p = 1/2$.
- The distribution of the "polarity" of literals[2] is uniform. That is, if r equals the probability that a literal is positive, previous analyses assumed $r = 1/2$.

The natural question arises: how does the distribution of $\mathrm{E}\left[X_f\right]$ change when $p \neq 1/2$ or $r \neq 1/2$?

Notice that in §1 we assumed that $P(Y_{ij} = 0) = 1/2$. In general, we know that $Y_{ij} = 0$ when the either the literal is positive and its variable is assigned `false` or the literal is negative and its variable is assigned `true`. That is,

$$P(Y_{ij}) = P\left((T \cap \neg S) \cup (\neg T \cap S)\right) = P(T)P(\neg S) + P(\neg T)P(S)$$

where T is the event that the variable is assigned `true`, and S is the event that the literal is positive. Then we know

$$P(Y_{ij}) = p(1-r) + (1-p)r = p + r - 2pr$$

and thus

$$\begin{aligned}\mathrm{E}\left[C_i \mid L_i = l_i\right] &= 1 - P\left(\bigcap_{j=1}^{l_i} Y_{ij} = 0\right) = 1 - \prod_{j=1}^{l_i} P(Y_{ij} = 0) \\ &= 1 - \prod_{j=1}^{l_i} (p + r - 2pr) = 1 - (p + r - 2pr)^{l_i}.\end{aligned}$$

[2] By the "polarity" of a literal we mean whether the literal is negated (which makes it negative) or not (which makes it positive).

Recalling the analysis for $\mathrm{E}[X_f]$ in §1,

$$\mathrm{E}[X_f] = \mathrm{E}\left[\frac{1}{m}\sum_{i=1}^{m}\mathrm{E}[C_i \mid L_i]\right] = \mathrm{E}\left[\frac{1}{m}\sum_{i=1}^{m}\left(1-(p+r-2pr)^{L_i}\right)\right]$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left(1-\mathrm{E}\left[(p+r-2pr)^{L_i}\right]\right) = \frac{1}{m}\sum_{i=1}^{m}\left(1-\sum_{j=1}^{n}\frac{1}{n}(p+r-2pr)^{j}\right)$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left(1-\frac{1}{n}(p+r-2pr)\left(\frac{1-(p+r-2pr)^{n}}{1-(p+r-2pr)}\right)\right)$$

$$= 1-\frac{1}{n}(p+r-2pr)\left(\frac{1-(p+r-2pr)^{n}}{1-(p+r-2pr)}\right).$$

Notice that this is a generalization for the case when $p = r = 1/2$, since

$$p+r-2pr = \frac{1}{2}+\frac{1}{2}-2\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 1-\frac{1}{2} = \frac{1}{2}$$

and thus

$$1-\frac{1}{n}(p+r-2pr)\left(\frac{1-(p+r-2pr)^{n}}{1-(p+r-2pr)}\right)$$
$$= 1-\frac{1}{n}\left(\frac{1}{2}\right)\left(\frac{1-1/2^{n}}{1-1/2}\right) = 1-\frac{1}{n}\left(1-1/2^{n}\right)$$

which matches the result from §1.

Again, if we assumed that the number of literals in each clause is fixed such that $L_i = l$ for all $L_i \in L$,

$$\mathrm{E}[X_f] = \mathrm{E}\left[\frac{1}{m}\sum_{i=1}^{m}\mathrm{E}[C_i \mid L_i = l]\right]$$
$$= \mathrm{E}\left[\frac{1}{m}\sum_{i=1}^{m}\left(1-(p+r-2pr)^{l}\right)\right] = 1-(p+r-2pr)^{l}.$$

And if we set $p = r = 1/2$,

$$\mathrm{E}[X_f] = 1-(p+r-2pr)^{l} = 1-\frac{1}{2^{l}}$$

which again matches the previous result.

What happens when we set only one of p or r to $1/2$? We get the remarkable result:

Theorem. When $p = 1/2$ (resp. $r = 1/2$) and $r \neq 1/2$ (resp. $p \neq 1/2$), then $\mathrm{E}[X_f]$ does not depend on r (resp. p).

Notice that if we let $p = 1/2$,

$$p + r - 2pr = \frac{1}{2} + r - 2\left(\frac{1}{2}\right)(r) = \frac{1}{2} + r - r = \frac{1}{2}$$

and the same result follows from similar reasoning when $r = 1/2$. Thus

$$\begin{aligned}\mathrm{E}[X_f] &= 1 - \frac{1}{n}(p + r - 2pr)\left(\frac{1 - (p + r - 2pr)^n}{1 - (p + r - 2pr)}\right) \\ &= 1 - \frac{1}{n}\left(\frac{1}{2}\right)\left(\frac{1 - 1/2^n}{1 - 1/2}\right) = 1 - \frac{1}{n}(1 - 1/2^n)\end{aligned}$$

which is the same result as when $p = r = 1/2$.

Intuitively, this means that if the KT algorithm uniformly assigned `true` or `false` to variables, the proportion of clauses satisfied *is independent of the proportion of literals that are positive/negative*. Conversely, if the polarity of literals is uniformly distributed over positive/negative, the distribution of clauses satisfied *is independent of the proportion of variables to which the KT algorithm assigns* `true`/`false`.

Example. Let $p = 1$, $r = 1/2$, and $n = 3$. Then

$$\mathrm{E}[X_f] = 1 - \frac{1}{3}\left(1 - 1/2^3\right) = 1 - \frac{1}{3}\left(\frac{7}{8}\right) = 1 - \frac{7}{24} = \frac{17}{24}.$$

Thus even when we assign `true` to *all* variables, truly the most trivial algorithm for MaxSAT imaginable, we still satisfy over 70% of clauses. This speaks to the ease with which SAT can be approximated, even though it is NP-complete. Compare this to the Traveling Salesman Problem, where it is both NP-complete and impossible to approximate (unless $P = NP$).

3 - Experiments

We ran experiments to empirically verify our analyses above. The experiments consisted of generating a CNF formula with 1000 clauses and running the KT

algorithm to generate a model for the formula[3]. We modified several parameters, including (1) the number of literals per clause (n), (2) whether the number of literals per clause is uniformly random (from 1 to n) or constant (always n), (3) the proportion of literals in the formula that are positive (r), and (4) the proportion of variables assigned `True` by the KT algorithm (p).

Random literals?	n	r	p	% sat	expected
Yes	3	0.5	0.5	0.7075	0.7083
Yes	3	0.3	0.5	0.6996	0.7083
Yes	5	0.5	0.5	0.8076	0.8063
Yes	5	0.3	0.8	0.7010	0.7036
No	3	0.5	0.5	0.8751	0.8750
No	3	0.5	1.0	0.8741	0.8750
No	5	0.5	0.5	0.9688	0.9688
No	5	0.4	0.5	0.9616	0.9688

We ran 100 trials for each parameter setting and calculated the average number of clauses satisfied in a formula. The table above summarizes our results. The first four columns specify the parameter setting for that experiment; the `% sat` column denotes the average proportion of clauses satisfied in a formula over 100 trials. The `expected` column denotes the predicted proportion of satisfied clauses by the analyses above. The results strongly indicate wide agreement between the empirically observed results and the predicted proportions, which vindicates the analyses above as correct.

[3]We implemented the experiments in Haskell. Go to github.com/rolph-recto/cs6161-project to see the code.